Solution Architecture

AI/ML and Generative AI Track • Project Documentation

Project	Trip Genius AI (VoyagePlan Travel Planner)	**Date**	26 October 2026
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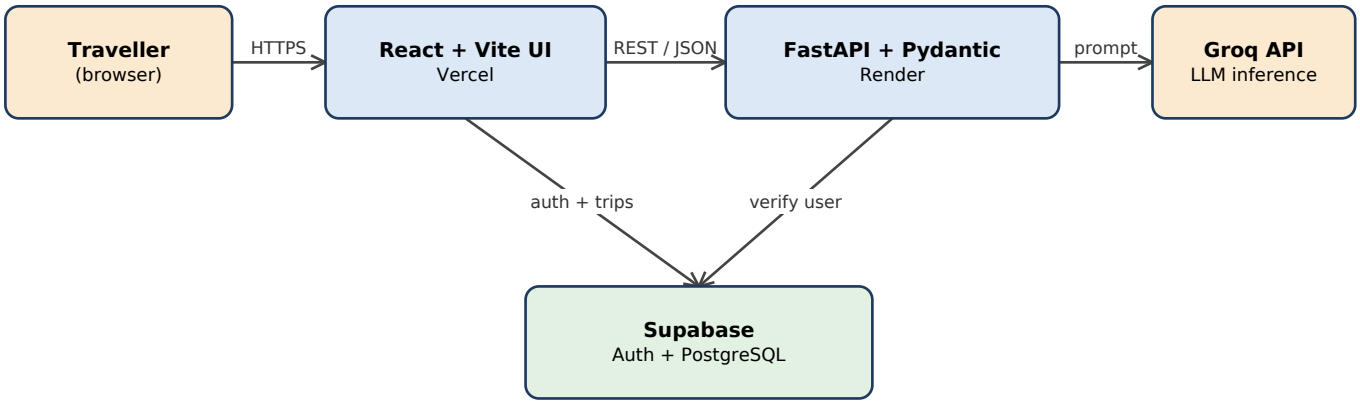
Architecture diagram



High-level architecture

Layer / component	Responsibilities	Technology
Presentation layer	Planner form, itinerary page, authentication screens, My Trips and navigation	React + Vite on Vercel
API communication	Send generation requests; handle loading, success and errors	Frontend API client over HTTPS
Application layer	Validate input, orchestrate generation and return a response	FastAPI, Python, Pydantic on Render
AI integration	Create itinerary content from preferences using server-side credentials	Groq API
Authentication	Sign-up, sign-in and session handling	Supabase Auth
Data layer	Persist user trips and retrieve owned records	Supabase PostgreSQL (RLS)
Hosting	Build the frontend and run the API service	Vercel + Render

Request flow

Traveller to React planner, then FastAPI request validation, then Groq, then response schema, then the frontend itinerary view. Authentication and saved-trip operations use Supabase and rely on a verified user identity rather than a client-supplied user ID.

Architecture constraints

- The browser must never receive server-side AI secrets.
- CORS allows only the approved frontend origin (the deployed Vercel URL).
- Frontend API base URL is held in VITE_API_BASE_URL; backend secrets live in Render environment variables.
- Add timeouts and readable errors for unavailable external services.